

Homework 4

Reinforcement Learning

PDF Submission

Exercise 2: Deep Q-Network (DQN)

Exercise 3: Proximal Policy Optimization (PPO)

Exercise 4: Soft Actor-Critic (SAC)

Due: 16.04.2026

Exercise 1

Dynamic Programming in GridWorld
(MDP — images for video reference)

Exercise 1 — Theoretical Questions

Q1: What is the difference between policy iteration and value iteration in terms of their update procedures?

Computationally, policy iteration computes $V(s)$ assuming we are following a certain policy (which we update separately), requiring a full convergence loop. Value iteration directly computes $V(s)$ as the maximum extractable value from each state by an argmax (one sweep), and computes a greedy policy from that. Both converge to the same optimal policies though.

Q2: What happens if the discount factor gamma is close to 0 or 1?

If gamma is close to zero essentially the policy will be very short-sighted in the sense that it will value only immediate rewards. If it's close to one on the other hand, future rewards will affect computations as much as immediate rewards, making the policy more long-sighted.

Q3: How does increasing the slip probability (slip_chance) affect the optimal policy? Compare slip_chance = 0.0, 0.01, and 0.2. Why does the agent behave more conservatively as stochasticity increases?

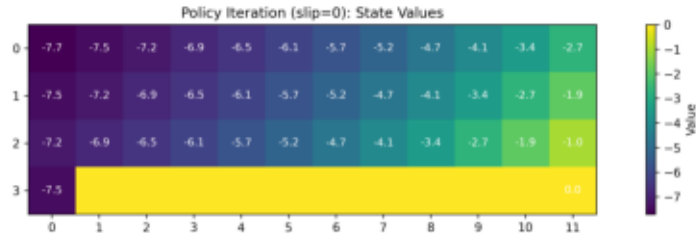
It makes the agent move away from the cliff (taking a longer route) more progressively as increasing slip chance makes the agent fall to the cliff once in a while if it stays close to it.

- slip_chance=0: agent is inclined to take the shortest path possible, moving close to the cliff.
- slip_chance=0.01: values for states close to the cliff get lower and the agent is somewhat motivated to move away.
- slip_chance=0.2: values for states close to the cliff get even lower and agent is much more inclined to move away.

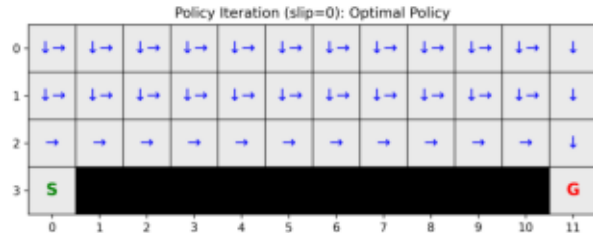
The agent behaves more conservatively as stochasticity increases precisely because the risk of falling from the cliff increases, which translates to lower state-values for states along the cliff (due to -100 reward) and an optimal policy that moves away from these states.

Policy Iteration — slip_chance = 0

State Value Function

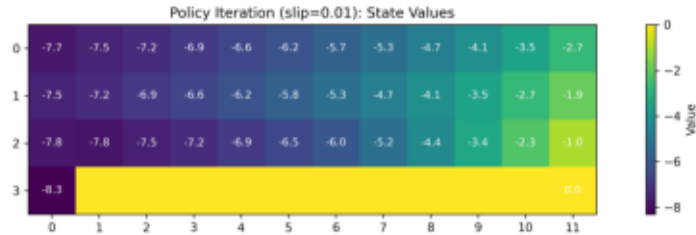


Optimal Policy

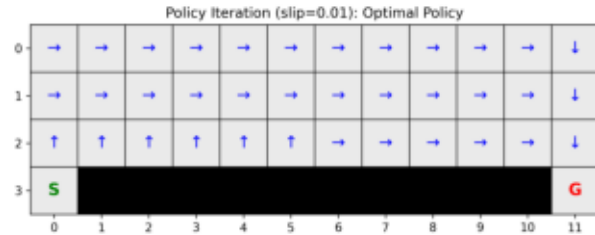


Policy Iteration — slip_chance = 0.01

State Value Function

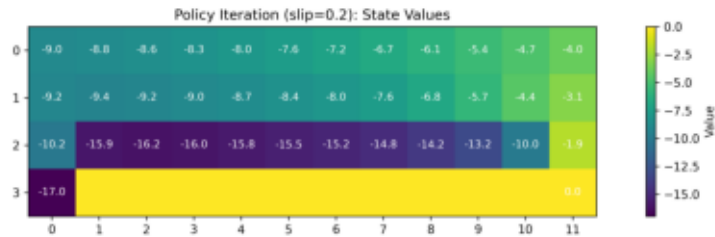


Optimal Policy

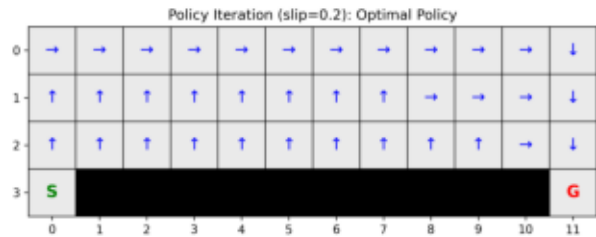


Policy Iteration — slip_chance = 0.2

State Value Function

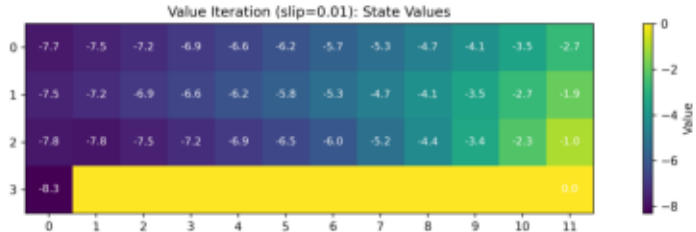


Optimal Policy

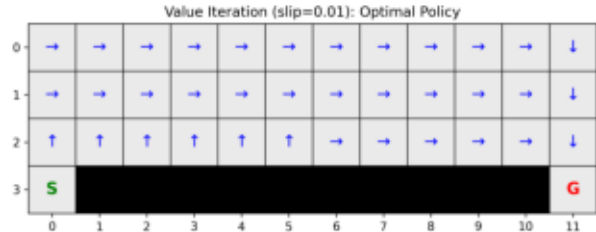


Value Iteration — slip_chance = 0.01

State Value Function

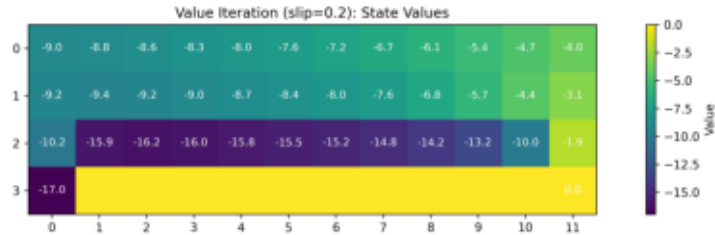


Optimal Policy

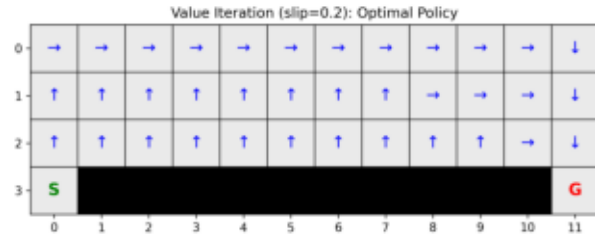


Value Iteration — slip_chance = 0.2

State Value Function



Optimal Policy



Exercise 2

Deep Q-Network (DQN) on CartPole

Exercise 2 — Theoretical Questions

Q1: Why is experience replay important in DQN?

To break temporal correlation (which breaks iid assumption of SGD) across training samples, sampling from different parts of different episodes is needed. Also it provides data efficiency.

Q2: What is the role of the target network in DQN? How does it improve stability?

It provides a "fixed target" for the DQN network to approximate, otherwise the network would be chasing a moving target, since every gradient update would change the target.

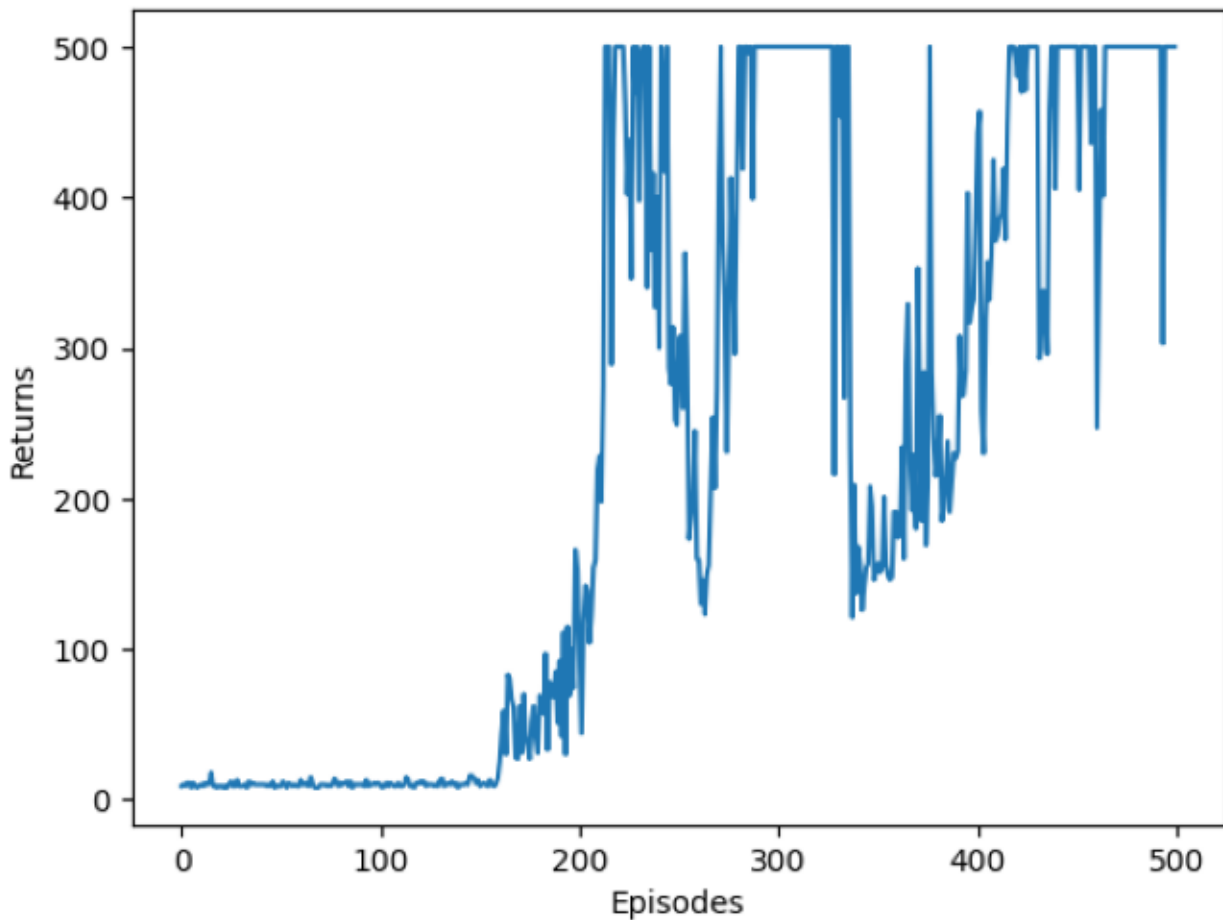
Q3: What is Double DQN, and how does it reduce overestimation bias compared to standard DQN?

It changes which network selects the best looking action (online) and which network evaluates how good that action actually is (target). In DQN both evaluation and selection are done by the online network which causes overestimation.

Exercise 2 — DQN Training Curve

DQN Training Curve

DQN on CartPole-v1



Exercise 2 — DQN Evaluation Summary

===== Evaluation Summary =====

Number of episodes : 50

Mean return : 500.00

Std return : 0.00

Min return : 500.00

Max return : 500.00

Median return : 500.00

Mean length : 500.00

Std length : 0.00

Success threshold : 475.0

Success rate : 100.0%

Exercise 3

Proximal Policy Optimization (PPO) on SO100

Exercise 3 — Theoretical Questions

Q1: Why does PPO clip the probability ratio instead of directly constraining the KL divergence like TRPO? What goes wrong if you remove clipping entirely?

TRPO enforces the KL constraint directly but requires expensive second-order optimization. PPO clips the ratio as a cheaper proxy. Without clipping, nothing stops the policy from taking huge destructive steps.

Q2: PPO throws away all collected data after each update. Why can't you simply reuse old rollouts for more gradient steps?

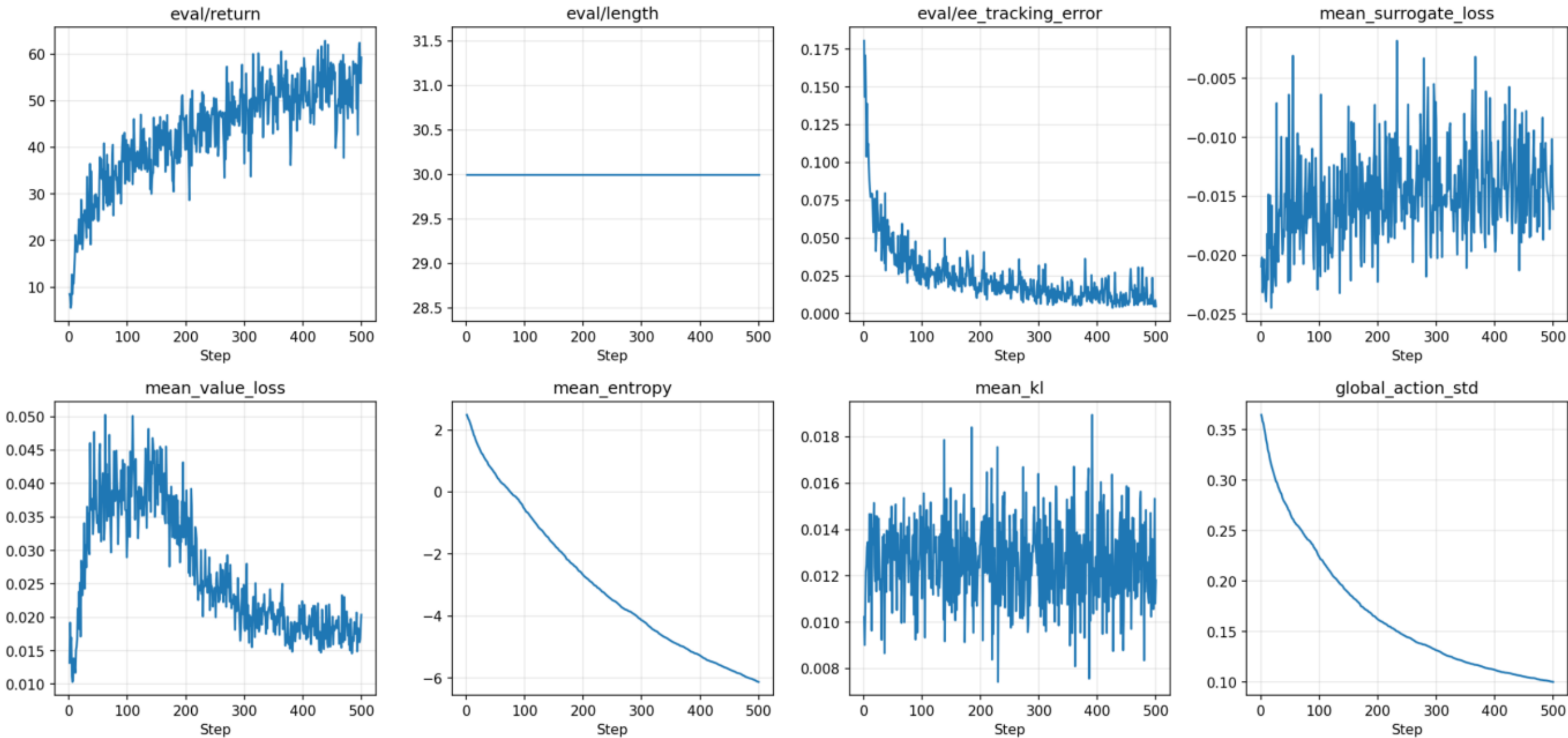
PPO is on-policy so advantage estimates assume data was collected under the current policy. After an update the policy has changed, making reuse biased.

Q3: What does the GAE parameter λ control? What happens at the extremes $\lambda = 0$ and $\lambda = 1$?

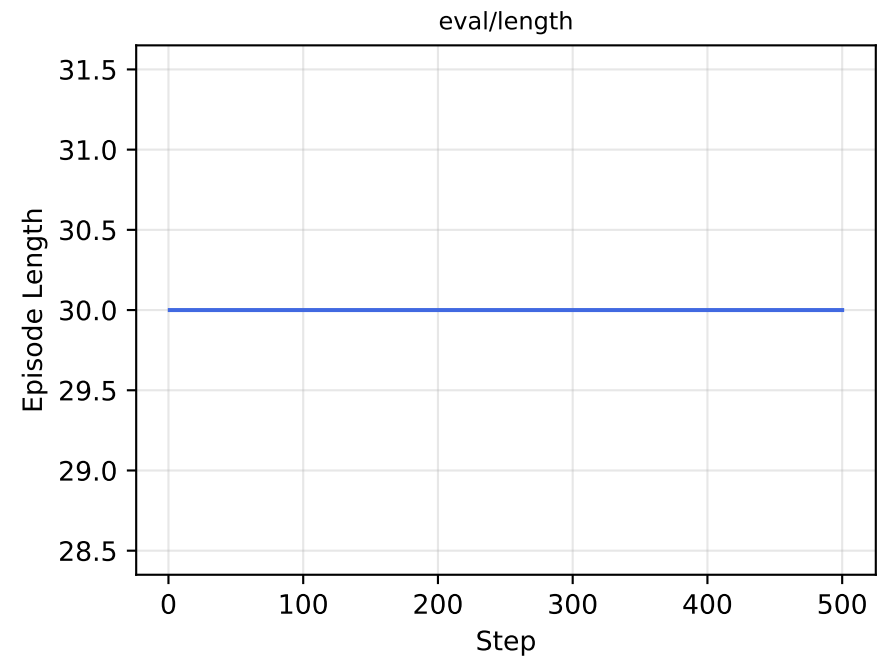
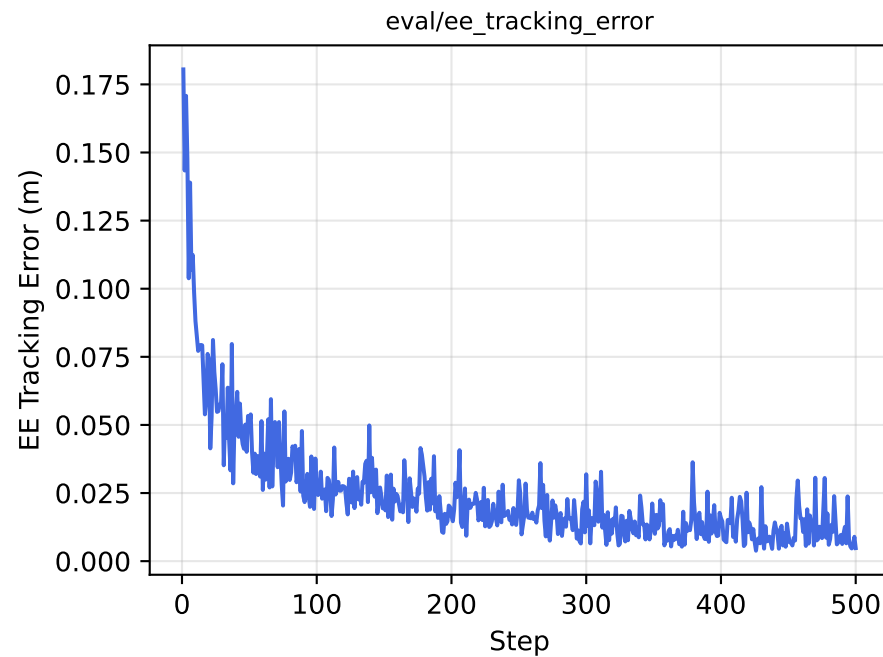
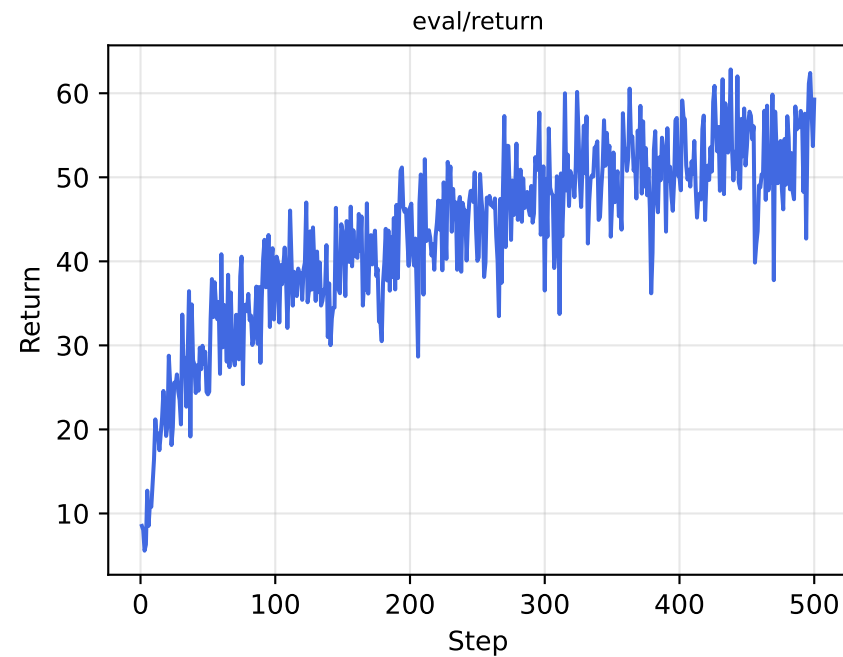
λ controls the bias-variance tradeoff in the advantage estimate. At $\lambda=0$ it's just the TD error (low variance, high bias), at $\lambda=1$ it's the full Monte Carlo return (unbiased, high variance).

Exercise 3 — PPO Training Curves (TensorBoard)

PPO Training Curves

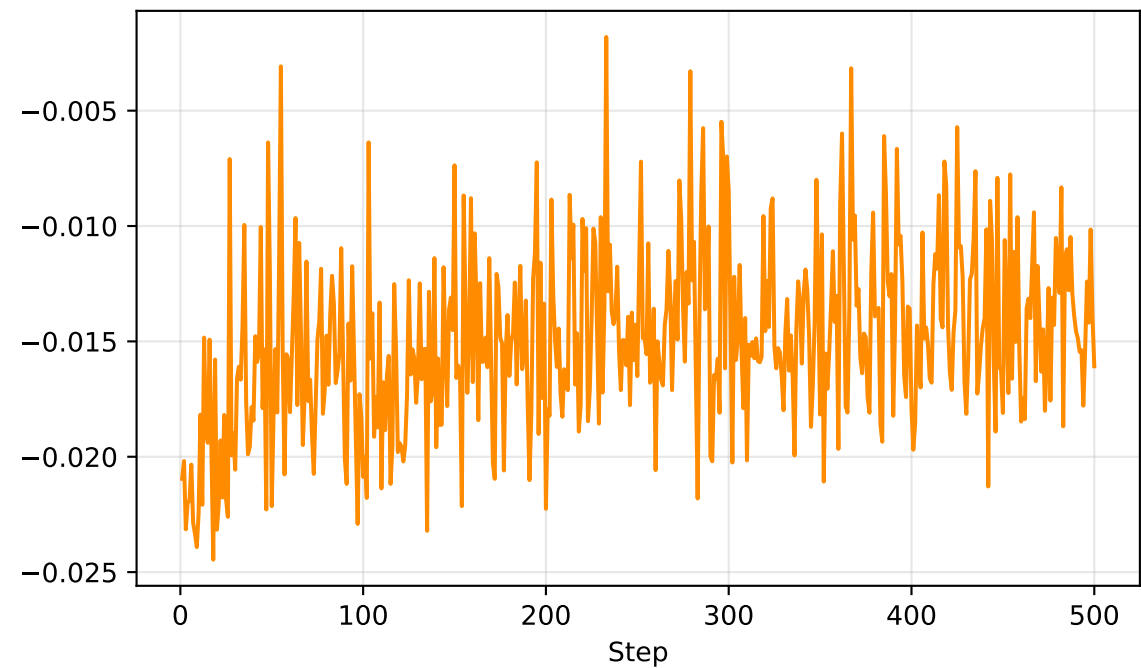


Exercise 3 — PPO Evaluation Metrics

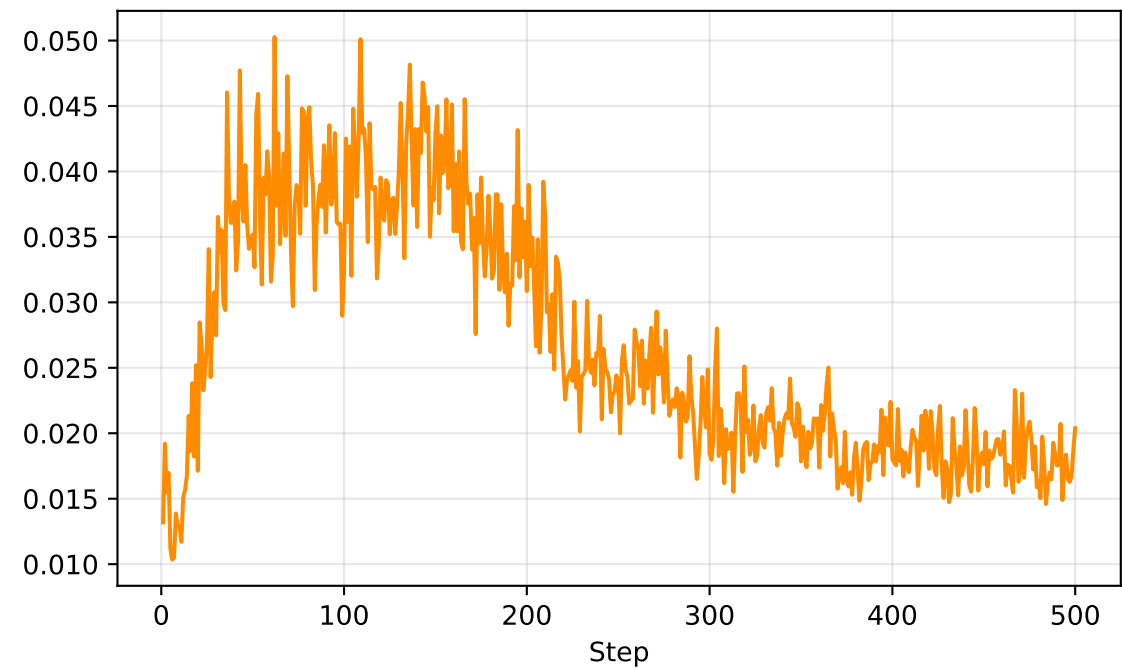


Exercise 3 — PPO Training Loss Curves

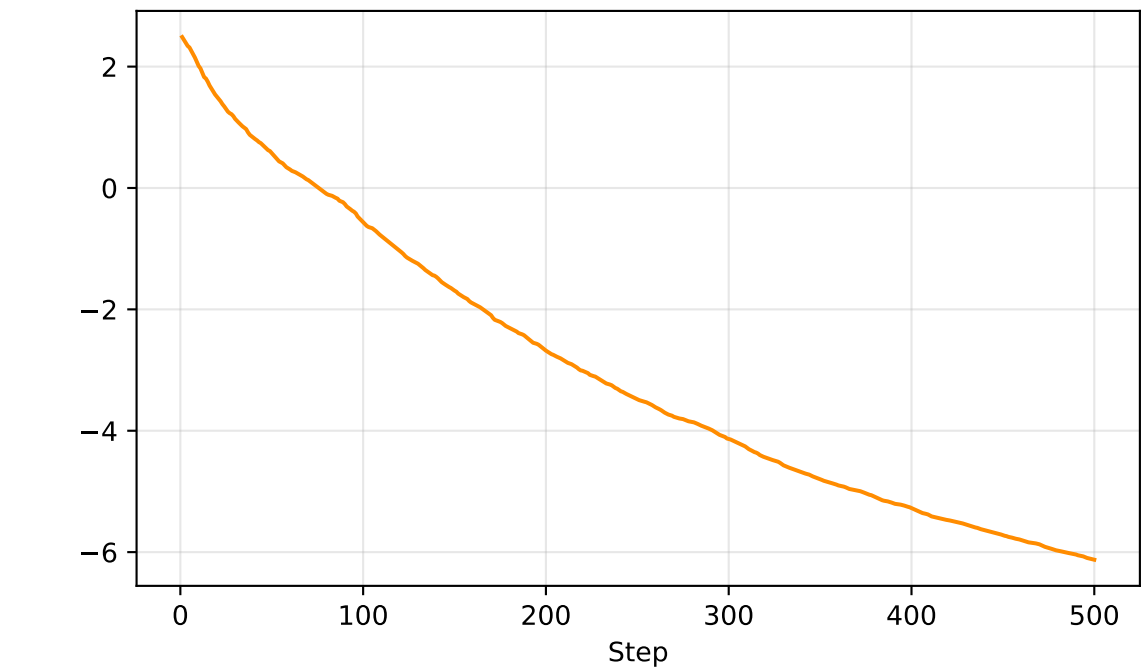
Surrogate Loss



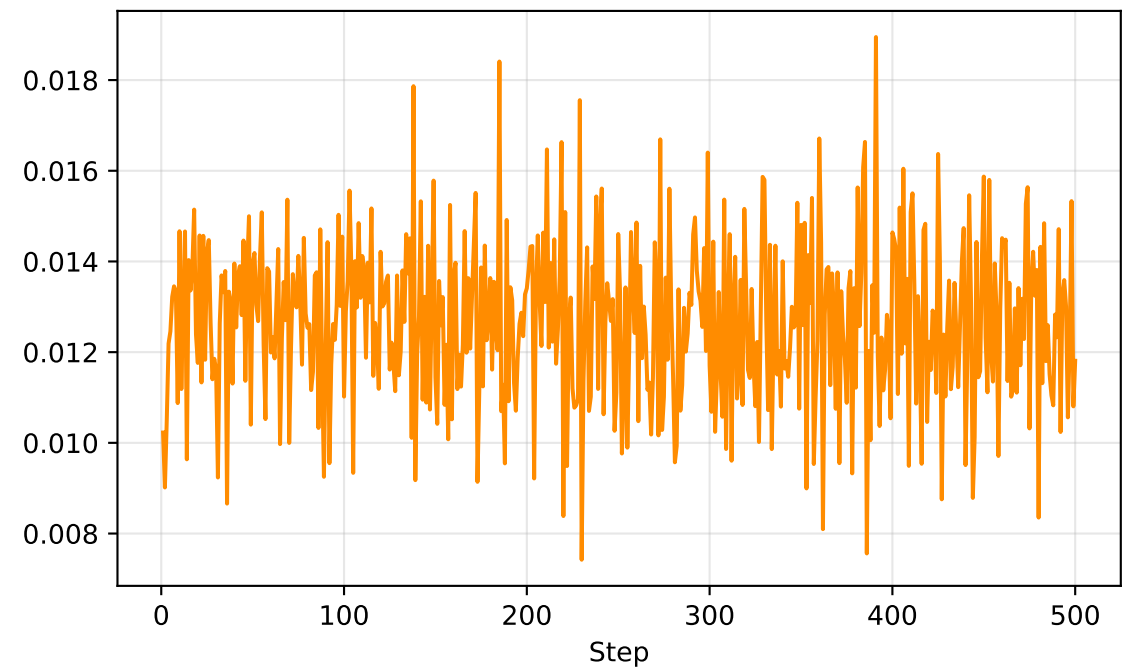
Value Loss



Entropy



Mean KL



Exercise 3 — PPO Evaluation Summary

==== Evaluation Summary ====

Number of episodes	: 20
Mean return	: 51.613
Std return	: 11.656
Min return	: 20.265
Max return	: 70.246
Median return	: 52.741
Mean length	: 30.00
Std length	: 0.00
Mean tracking error	: 0.018817
Std tracking error	: 0.015991
Min tracking error	: 0.004162
Max tracking error	: 0.071369

Exercise 4

Soft Actor-Critic (SAC) on SO100

Exercise 4 — Theoretical Questions

Q1: SAC adds an entropy bonus to the reward. What are the benefits of this?

It encourages exploration and prevents the policy from collapsing to a suboptimal deterministic solution too early.

Q2: SAC squashes actions through tanh. Why does this require a log-probability correction?

tanh is nonlinear so the Gaussian log-prob of the pre-squashed sample doesn't match the log-prob of the output. You need to subtract the log-Jacobian to correct for it.

Q3: The temperature α is tuned automatically. What happens when the policy's entropy is above vs. below the target?

If entropy is above target α decreases to make the policy more focused, if it's below α increases to push toward more exploration.

Q4: How does SAC compare with PPO in terms of update-to-data (UTD) ratio? (UTD = gradient update steps / environment steps)

SAC has a much higher UTD ratio since it's off-policy and replays transitions many times from the buffer, whereas PPO discards rollouts after each update.

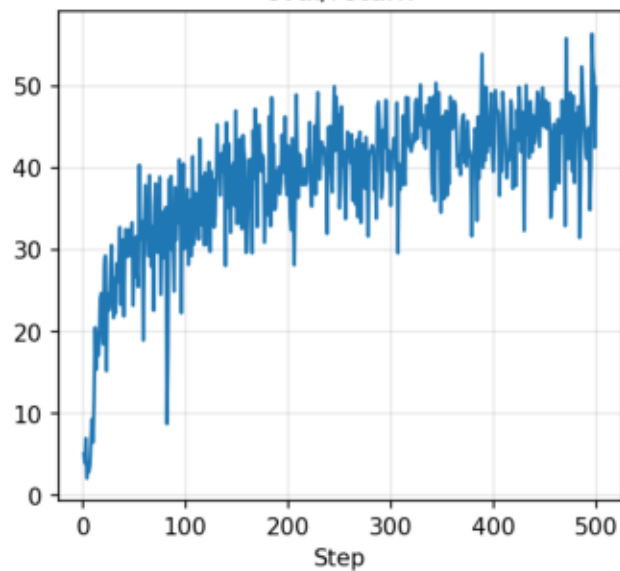
Q5: Briefly discuss the advantages and disadvantages of on-policy vs. off-policy algorithms.

On-policy algorithms like PPO are simpler and more stable but sample-inefficient. Off-policy ones like SAC are much more sample-efficient by reusing old data, but more complex and potentially less stable.

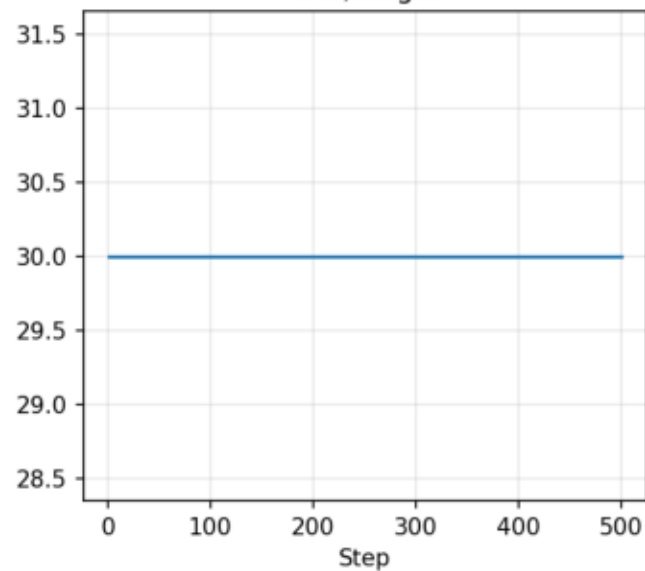
Exercise 4 — SAC Training Curves (TensorBoard)

SAC Training Curves

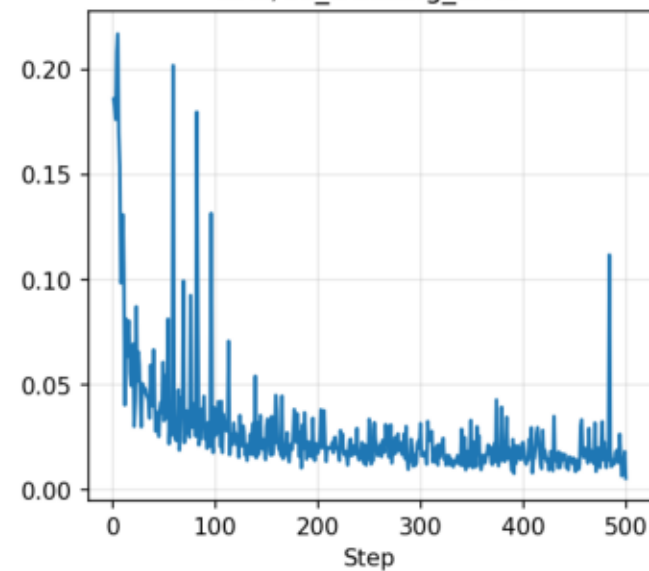
eval/return



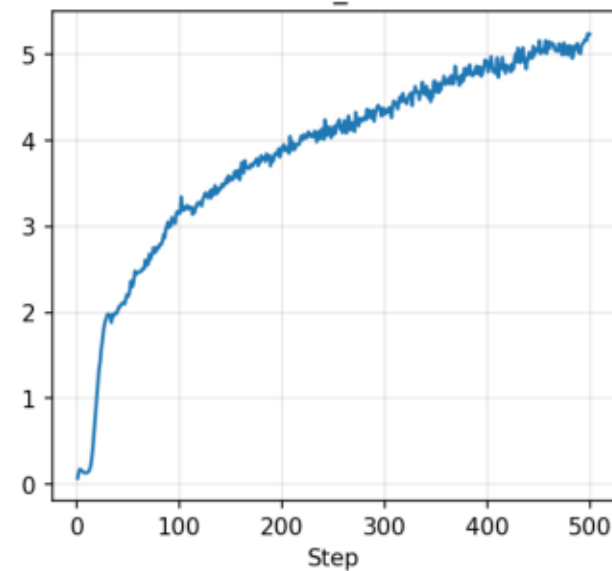
eval/length



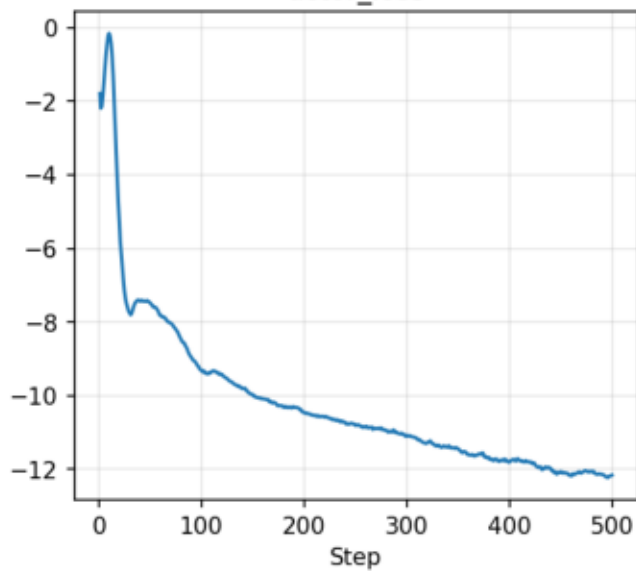
eval/ee_tracking_error



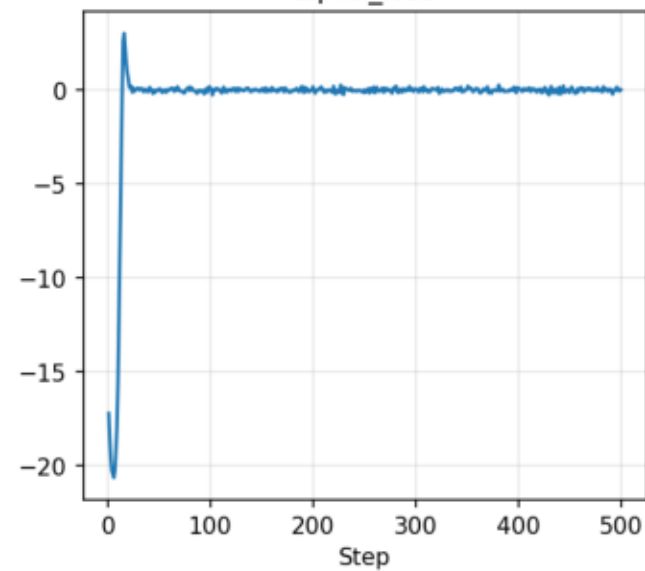
critic_loss



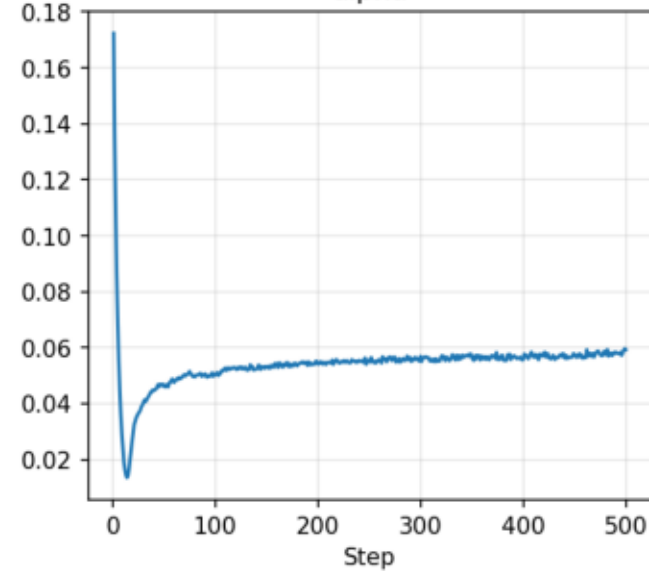
actor_loss



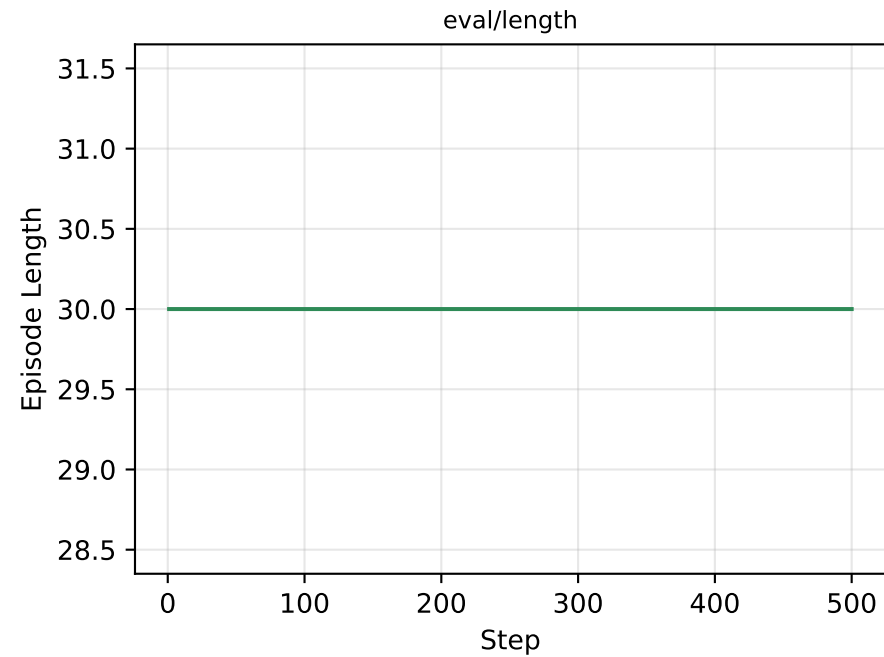
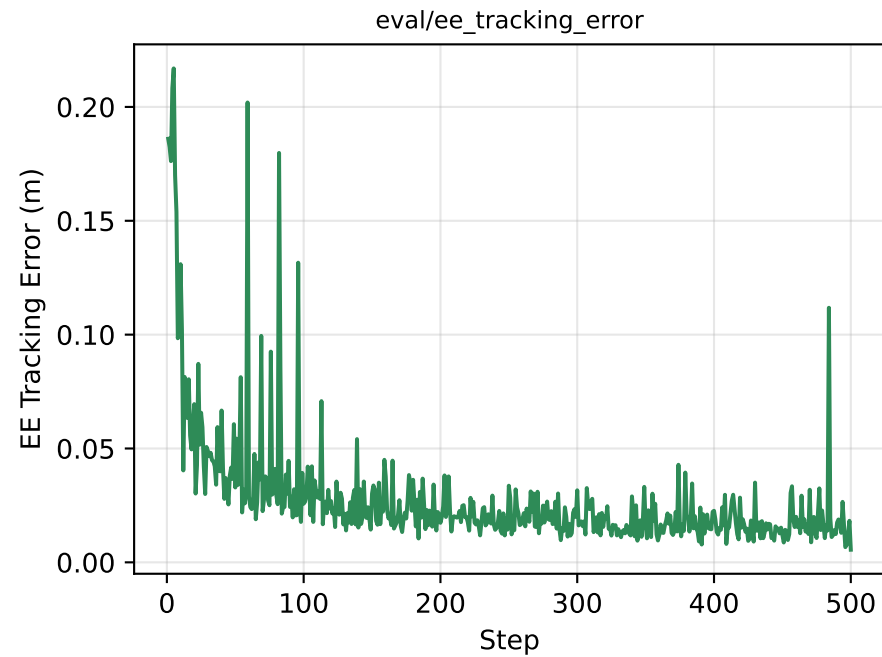
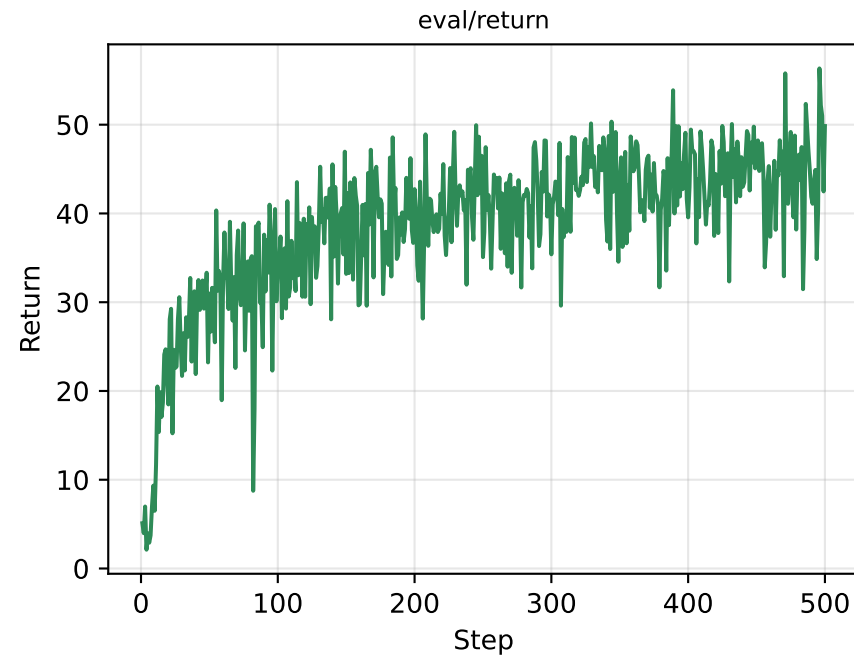
alpha_loss



alpha

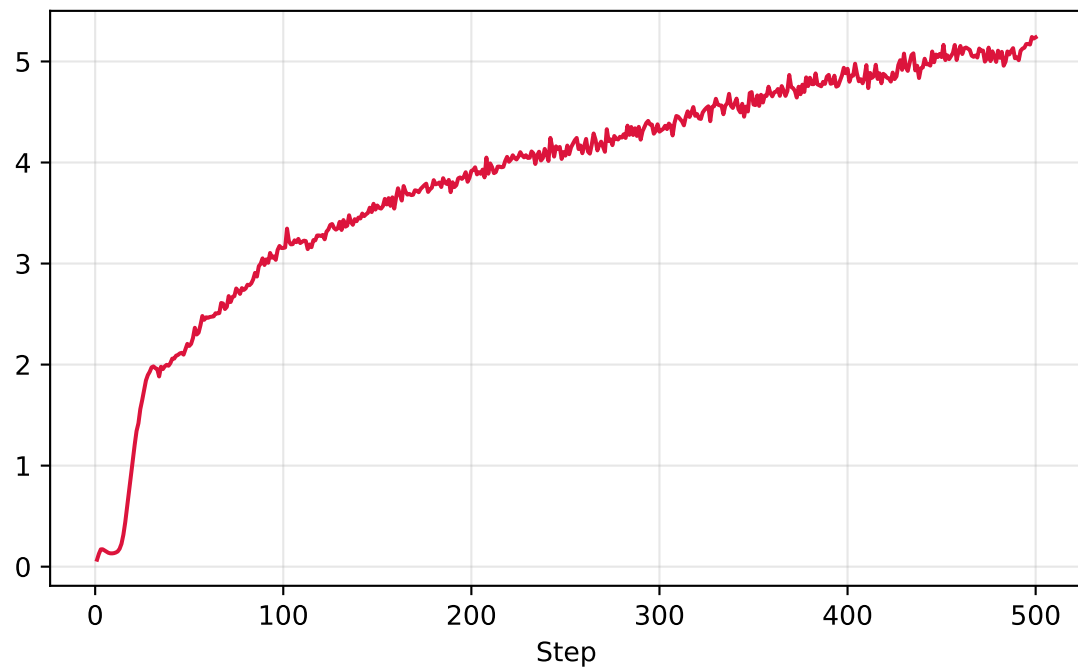


Exercise 4 — SAC Evaluation Metrics

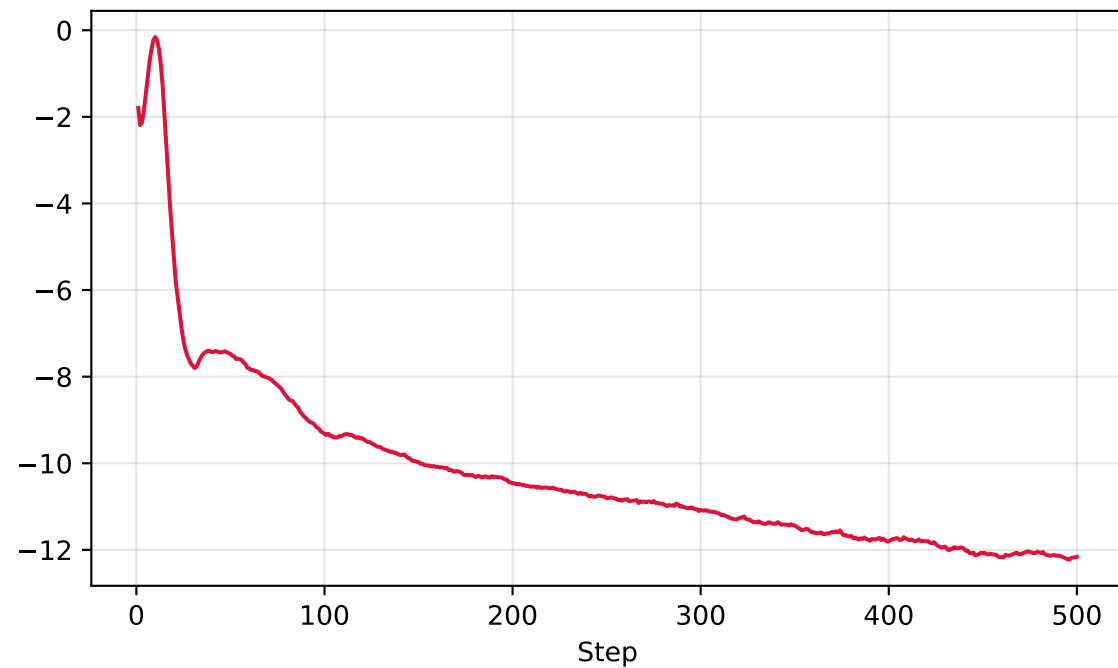


Exercise 4 — SAC Training Loss Curves

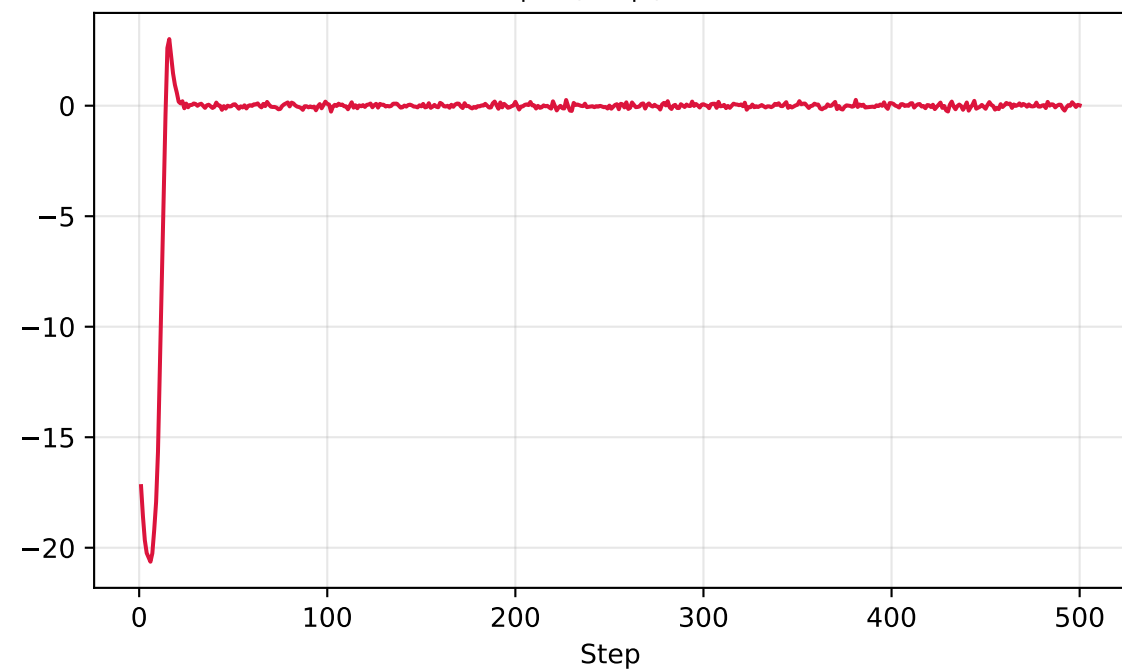
Critic Loss



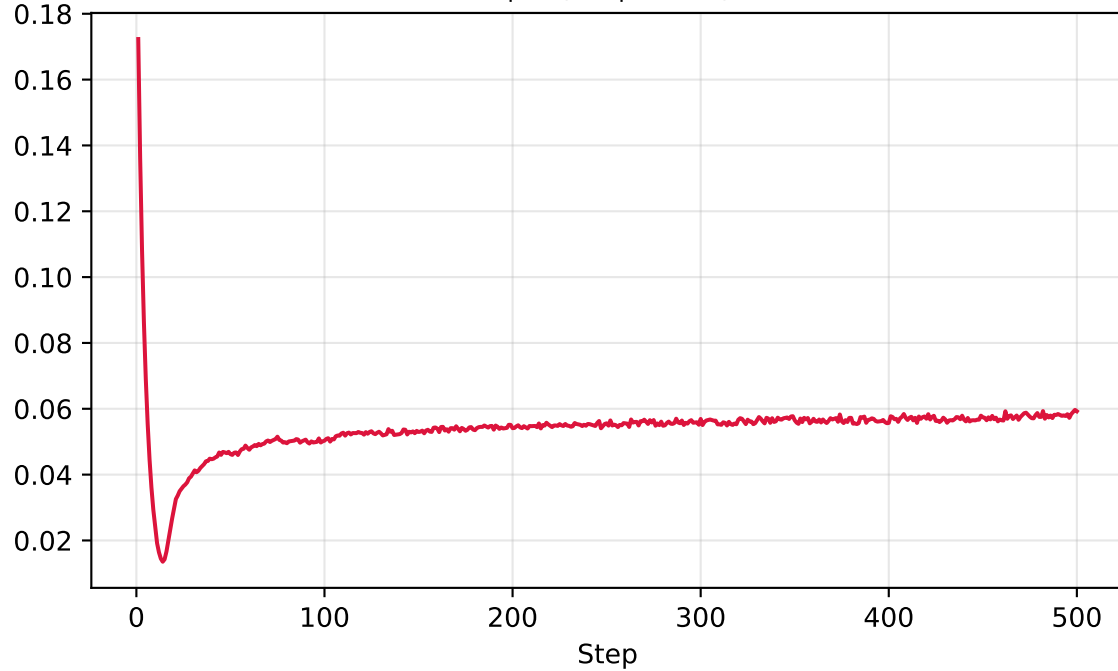
Actor Loss



Alpha (Temp.) Loss



Alpha (Temperature)



Exercise 4 — SAC Evaluation Summary

```
===== Evaluation Summary =====  
Number of episodes      : 20  
Mean return             : 47.168  
Std return              : 10.345  
Min return              : 20.932  
Max return              : 67.170  
Median return           : 49.721  
Mean length             : 30.00  
Std length              : 0.00  
Mean tracking error      : 0.024551  
Std tracking error       : 0.016066  
Min tracking error       : 0.007176  
Max tracking error       : 0.067506
```

PPO vs SAC — Side-by-Side Comparison

